Mathematics: analysis and approaches
Standard level
Paper 1
Topic 4 Statistics — box-and-whisker diagrams

5 November 2026

W4ss Academy practice set 1 | **Topic** Topic 4 Statistics — box-and-whisker diagrams | **Ref.** W4SS–MAA–QB–P1–01

Student name

50 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- Section A: answer all questions. Answers must be written within the answer boxes provided.
- Section B: answer all questions on the answer pages at the end of this paper.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[44 marks]**.

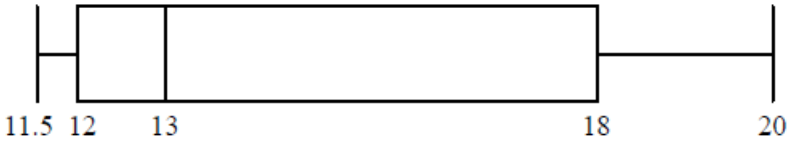
Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

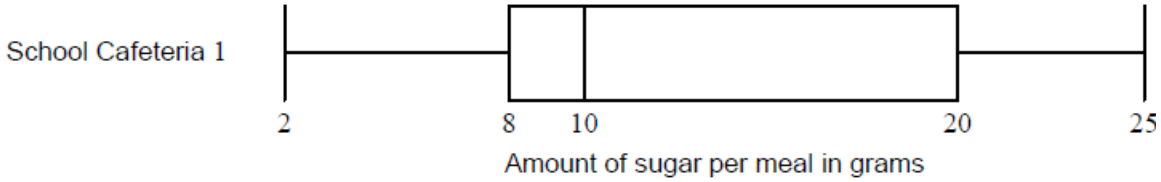
1. [Maximum mark: 2]

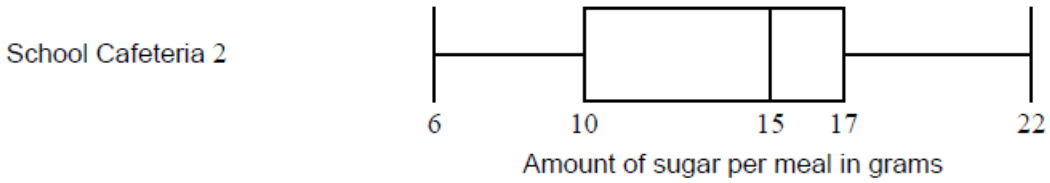
A health inspector analysed the amount of sugar in 500 different snacks prepared in various school cafeterias. The collected data are shown in the following box-and-whisker diagram.



Amount of sugar per snack in grams

The health inspector visits two school cafeterias. She inspects the same number of meals at each cafeteria. The data is shown in the following box-and-whisker diagrams.





Meals prepared in the school cafeterias are required to have less than 10 grams of sugar.

State, giving a reason, which school cafeteria has more meals that do not meet the requirement.

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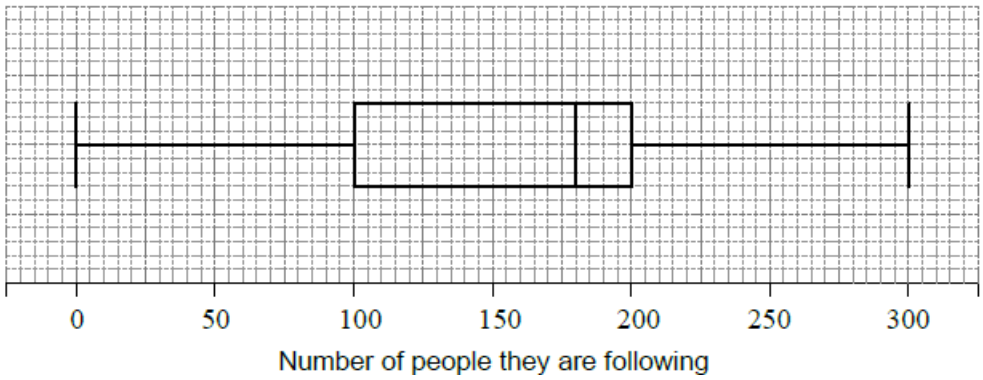
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2. [Maximum mark: 1]

In a high school, 160 students completed a questionnaire which asked for the number of people they are following on a social media website. The results were recorded in the following box-and-whisker diagram.



The following incomplete table shows the distribution of the responses from these 160 students.

Number of people they are following (x)	Number of high school students
$0 \leq x \leq 50$	4
$50 < x \leq 100$	
$100 < x \leq 150$	34
$150 < x \leq 200$	46
$200 < x \leq 250$	
$250 < x \leq 300$	16

Write down the mid-interval value for the $100 < x \leq 150$ group.

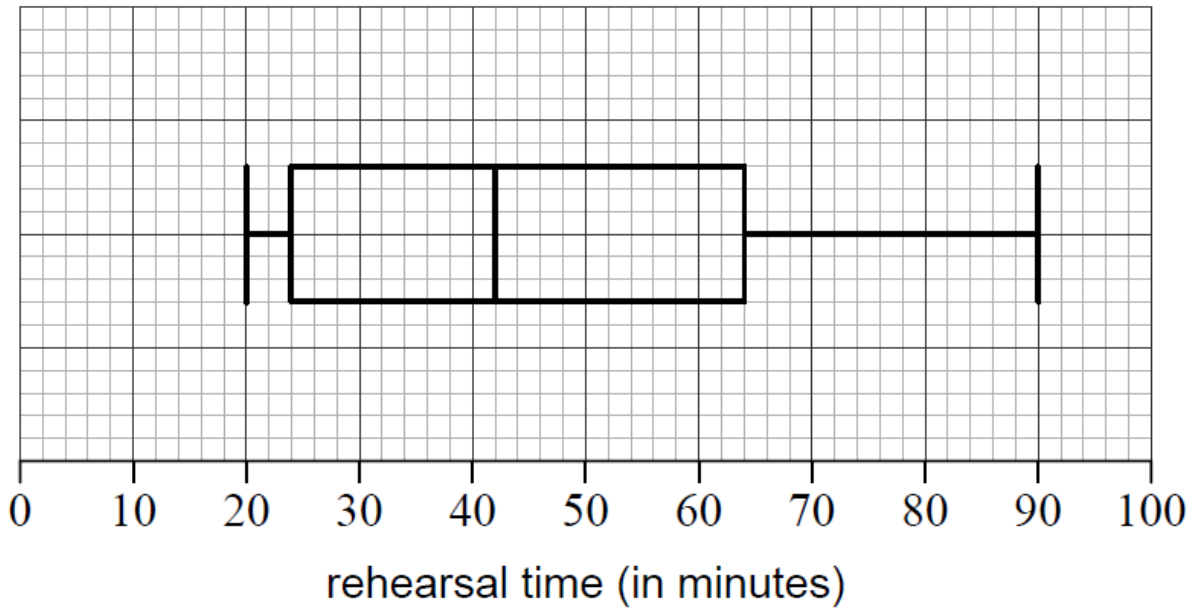
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3. [Maximum mark: 6]

Stephen was invited to perform a piano recital. In preparation for the event, Stephen recorded the amount of time, in minutes, that he rehearsed each day for the piano recital.

Stephen rehearsed for 32 days and data for all these days is displayed in the following box-and-whisker diagram.



(a) Write down the median rehearsal time. [1]

Stephen states that he rehearsed on each of the 32 days.

(b) State whether Stephen is correct. Give a reason for your answer. [2]

(c) On k days, Stephen practiced exactly 24 minutes. Find the possible values of k. [3]

(This question continues on the following page)

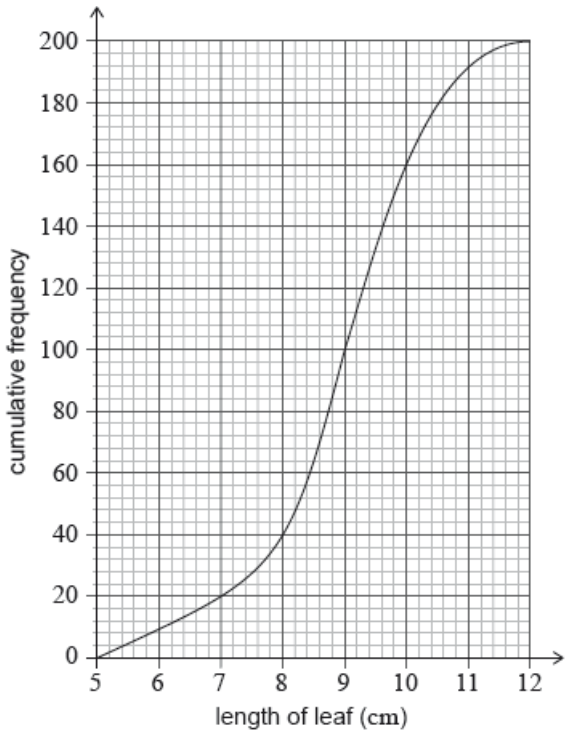
(Question 3 continued)

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4. [Maximum mark: 2]

For a study, a researcher collected 200 leaves from oak trees. After measuring the lengths of the leaves, in cm, she produced the following cumulative frequency graph.



(a) Write down the median length of these leaves. [1]

(b) Write down the number of leaves with a length less than or equal to 8 cm. [1]

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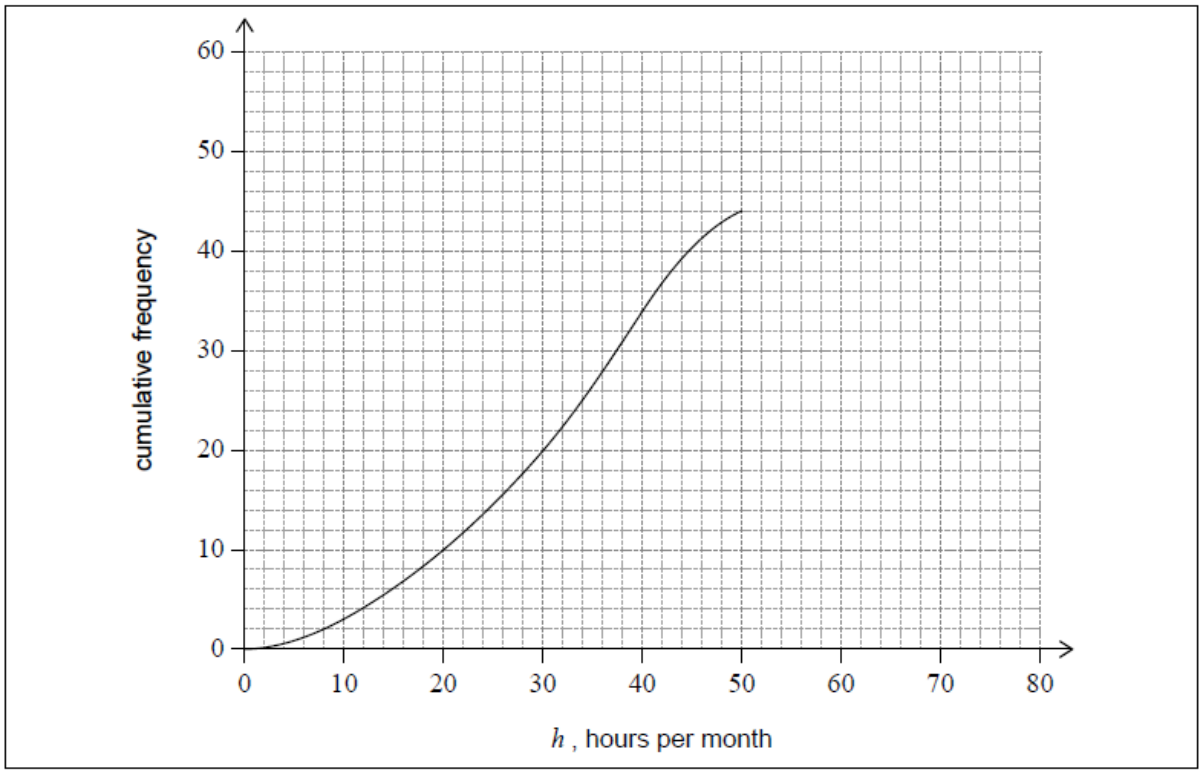
5. [Maximum mark: 6]

University students were surveyed and asked how many hours, h, they worked each month. The results are shown in the following table.

Hours per month, h	Frequency	Cumulative frequency
$0 < h \leq 10$	3	3
$10 < h \leq 20$	7	10
$20 < h \leq 30$	10	20
$30 < h \leq 40$	14	34
$40 < h \leq 50$	p	44
$50 < h \leq 60$	6	50
$60 < h \leq 70$	4	54
$70 < h \leq 80$	2	q

Use the table to find the following values.

The first five class intervals, indicated in the table, have been used to draw part of a cumulative frequency curve as shown.



(a) (i) p. [1]

(ii) q. [1]

(This question continues on the following page)

(Question 5 continued)

(b) On the same grid, complete the cumulative frequency curve for these data. [2]

(c) Use the cumulative frequency curve to find an estimate for the number of students who worked at most 35 hours per month. [2]

6. [Maximum mark: 4]

A group of 20 students travelled to a gymnastics tournament together. Their ages, in years, are given in the following table.

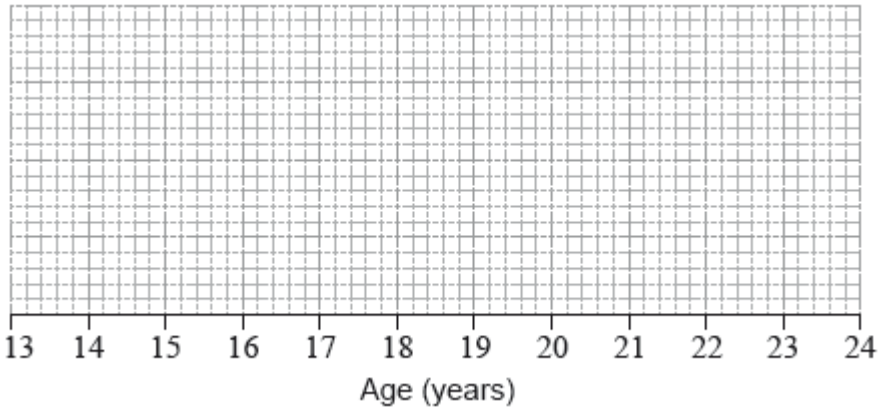
Age (years)	14	15	16	17	18	19	20	22
Frequency	1	2	7	1	4	1	1	3

The lower quartile of the ages is 16 and the upper quartile is 18.5.

(a) For the students in this group write down the median age. [1]

(Question 6 continued)

(b) Draw a box-and-whisker diagram, for these students' ages, on the following grid.



[3]

Section B

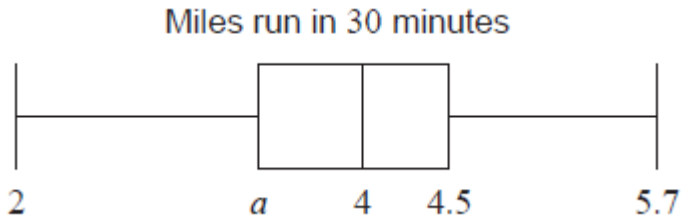
Answer **all** questions on the answer pages at the end of this paper. Please start each question on a new page.

7. [Maximum mark: 15]

Each athlete on a running team recorded the distance (M miles) they ran in 30 minutes.

The median distance is 4 miles and the interquartile range is 1.1 miles.

This information is shown in the following box-and-whisker plot.



(a) Find the value of a. [2]

The distance in miles, M, can be converted to the distance in kilometres, K, using the formula $K = \frac{8}{5}M$.

(b) Write down the value of the median distance in kilometres (km). [1]

The variance of the distances run by the athletes is $\frac{16}{9}$ km^2.

The standard deviation of the distances is b miles.

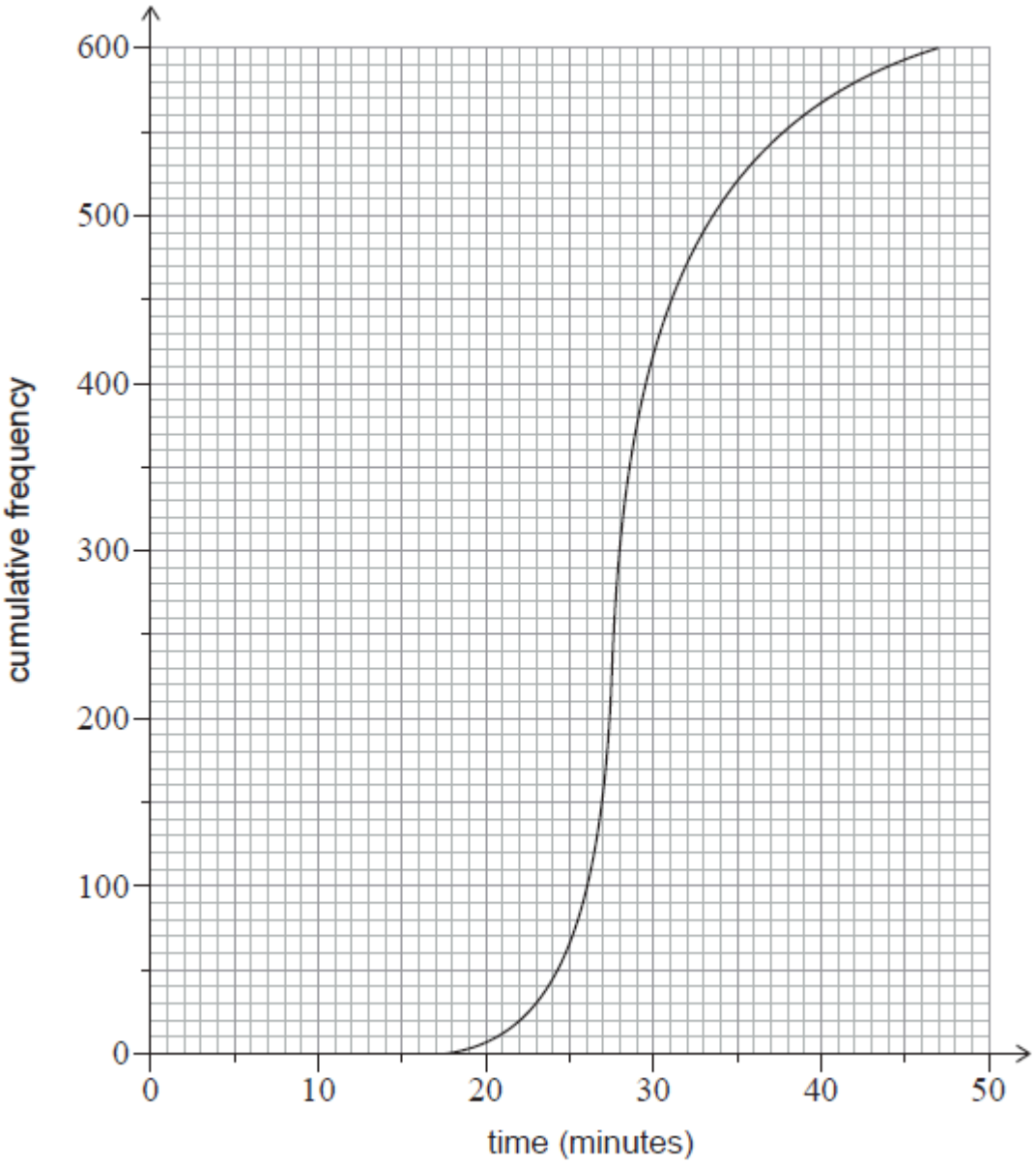
(c) Find the value of b. [4]

(This question continues on the following page)

Do not write solutions on this page.

(Question 7 continued)

A total of 600 athletes from different teams compete in a 5 km race. The times the 600 athletes took to run the 5 km race are shown in the following cumulative frequency graph.



There were 400 athletes who took between 22 and m minutes to complete the 5 km race.

(d) Find m. [3]

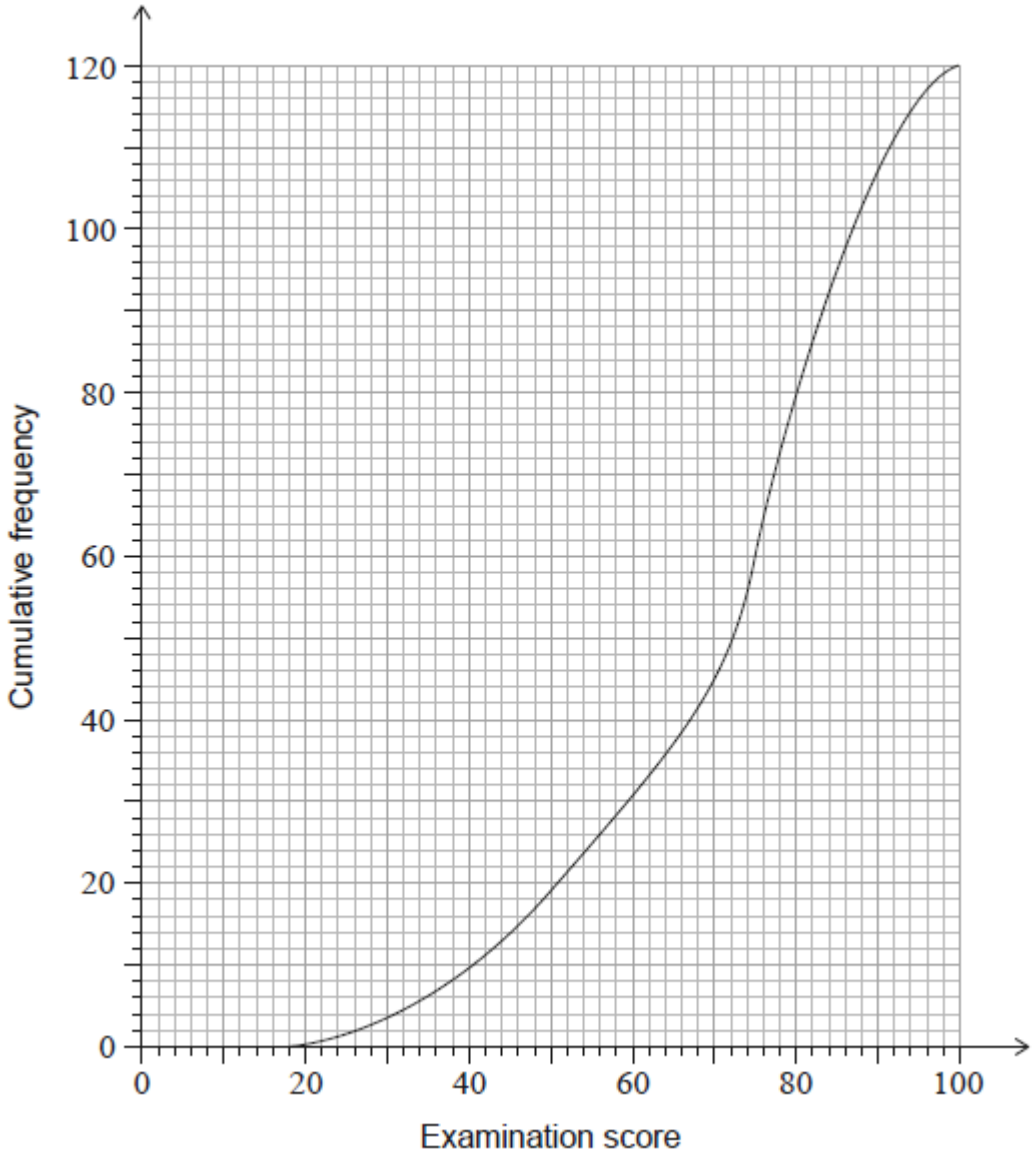
(e) The first 150 athletes that completed the race won a prize.

Given that an athlete took between 22 and m minutes to complete the 5 km race, calculate the probability that they won a prize. [5]

Do not write solutions on this page.

8. [Maximum mark: 8]

A group of 120 students sat a history exam. The cumulative frequency graph shows the scores obtained by the students.



The students were awarded a grade from 1 to 5, depending on the score obtained in the exam. The number of students receiving each grade is shown in the following table.

Grade	1	2	3	4	5
Number of students	6	13	26	a	b

The mean grade for these students is 3.65.

(a) Find the median of the scores obtained. [1]

(b) Find an expression for a in terms of b. [2]

(This question continues on the following page) **Turn over**

(Question 8 continued)

(c) (i) Find the number of students who obtained a grade 5. [3]

(ii) Find the minimum score needed to obtain a grade 5. [2]

Do not write solutions on this page.

Section B — answers to question 7

Section B — answers to question 7 (continued)

Section B — answers to question 7 (continued)

Section B — answers to question 8

Section B — answers to question 8 (continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability), as chosen in the class QuestionBank test. Some of them come from former courses (Mathematics SL, Mathematical studies SL). They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	19M.1.SL.TZ2.T_6 (former Mathematical studies SL course) · 4.2 comparing box-and-whisker diagrams
Section A — question 2	18M.1.SL.TZ1.T_6 (former Mathematical studies SL course) · 4.2 box-and-whisker diagram, grouped data
Section A — question 3	19N.1.SL.TZ0.T_15 (former Mathematical studies SL course) · 4.2, 4.3 box-and-whisker diagram, median, quartiles
Section A — question 4	17M.1.SL.TZ2.T_6 (former Mathematical studies SL course) · 4.2 cumulative frequency graph, median
Section A — question 5	19M.1.SL.TZ2.T_12 (former Mathematical studies SL course) · 4.2 cumulative frequency table and curve
Section A — question 6	17N.1.SL.TZ0.T_1 (former Mathematical studies SL course) · 4.2, 4.3 median, box-and-whisker diagram — part (a)(ii) of the original, relabelled (a): part (a)(i) is not in the test
Section B — question 7	20N.1.SL.TZ0.S_8 (former Mathematics SL course) · 4.2 box-and-whisker plot, cumulative frequency · 4.3 standard deviation · 4.6 conditional probability
Section B — question 8	21N.1.SL.TZ0.9 · 4.2 cumulative frequency graph · 4.3 mean of grouped data